

5. The dynamics

The next step is to construct a model for the dynamics. The idea, which essentially goes back to Boltzmann (see [12]), is that any given macro-state at any given time is extremely likely to develop into a state with higher entropy at the next moment of time, simply because there are so many more states with higher entropy than with lower entropy (compare with (3)). The problem with this in the present situation, however, is that this way of thinking in fact presupposes a preferred direction of time. Otherwise, given that the dynamical laws are time symmetric, why can we not similarly argue that the entropy should also grow when we go backward in time? (compare [9]).

There have been many attempts to avoid this problem by looking for defects in the symmetries. But my conclusion here is that we must actually accept Boltzmann's argument in both directions of time and hence we are led to the following:

Principle 1. At every moment of time t and for every state with entropy S , there are very many “accessible states” with higher entropy, both at the previous moment of time $t - 1$ and at the next one $t + 1$. On the other hand, the chance for finding such accessible states with lower entropy, both at times $t - 1$ and $t + 1$, is extremely small.

This principle also implies a shift of perspective in the search for time's arrow. Rather than trying to find the reason for the asymmetry, we must concentrate on understanding why we cannot observe the symmetric structure of the multiverse as a whole.

As still one more simplification, let us assume that the entropy can only change by ± 1 during each unit of time. This assumption, however, has to be modified near the endpoints (BB and BC) for the following reason: it is a very important aspect of this approach to assume that physics during the first and last moments is very different from the rest of the time, since at these moments quantum phenomena can be expected to become global. To model this in a simple way, we can split the life-span of our multiverse up into three parts:

$$[-T_0, -T_1] \cup [-T_1, T_1] \cup [T_1, T_0]. \quad (4)$$

Here the first and last parts may be called “the extreme phases,” which are characterized by the property that transition between very different states can be possible. During the “normal phase” in between on the other hand, physics is supposed to behave more or less as we are used to.

6. Modeling the dynamics

To construct a miniature multiverse for computational purposes, one can proceed as follows: first of all, in the very small multiverses studied here, the extreme phases will only last for one single unit of time. Also, for ease of notation, let us put $T_1 = m$, so that the moments of time can in this context be denoted as

$$-m - 1, -m, -m + 1, \dots, m - 1, m, m + 1. \quad (5)$$

The dynamics is specified by randomly choosing for each state at time t with entropy S , K edges to states at time $t + 1$ with entropy $S + 1$, and similarly K edges to states at time $t - 1$ with entropy $S + 1$ (with obvious modifications at the end-points). In this section, again to make everything as simple as possible, K will be set equal to 2. These random choices are in practice carried out by the random number